

Table of Contents:

Lecture 1	9
Introduction	9
1.1 Charles Babbage (1791-1871)	9
1.2 The Analytical Engine.....	9
1.3 Ada, Countess of Lovelace(1815-52)	9
1.4 Course Contents & Structure	10
Lecture 2	13
Evolution of Computing	13
2.1 Turing Machine – 1936.....	13
2.2 The “Turing test”	13
2.3 Vacuum Tube – 1904:.....	13
2.4 ABC – 1939	14
2.5 Harvard Mark 1 – 1943:.....	14
2.6 ENIAC – 1946:.....	14
2.7 Transistor – 1947	14
2.8 Floppy Disk – 1950.....	14
2.9 UNIVAC 1 – 1951	14
2.10 Compiler - 1952.....	15
2.11 ARPANET – 1969	15
2.12 Intel 4004 – 1971	15
2.13 Altair 8800 – 1975.....	15
2.14 Cray 1 – 1976	16
2.15 IBM PC – 1981	16
2.16 Apple Macintosh – 1984.....	16
2.17 World Wide Web -1989	16
2.18 Quantum Computing with Molecules	17
Lecture 3	18
3.1 Browser	18
3.2 URL.....	18
3.3 What is a Web site?.....	18
3.4 What is Home Page of a web site?	19
3.5 Who invented the Web & Why?	19
3.6 Future of the Web: Semantic Web.....	20
3.7 Useful Web page	20
Lecture 4	21
4.1 Computer Types According to Capability.....	21
4.2 Supercomputers	21
4.3 Mainframe Computers.....	21
4.4 Servers / Minicomputers	21
4.5 Desktops.....	21
4.6 Portables.....	21
4.7 Ranking w.r.t. installed number.....	22
4.8 All computers have the following essential hardware components:.....	22
4.9 Input Devices.....	23
4.10 What is Port?	24
4.11Many Types of Ports	24
4.12 Processor.....	24
4.13 Memory/Storage	24
4.14 Classifying Memory/Storage.....	25
4.15 Output Devices.....	25
4.16 Modem.....	25

Lecture 5.....	27
5.1 PC Parts.....	27
5.2 Inside of the CPU	27
5.3 The Processor Module	27
Lecture 6.....	28
6.1 To develop your personal Web page	28
Lecture 7.....	31
7.1 Microprocessor	31
7.2 Integrated Circuits.....	31
7.3 Devices.....	31
7.4 Microprocessor system.....	33
7.5 Micro-controllers	33
7.6 The Main Memory Bottleneck	33
7.7 Cache	33
7.8 Microprocessors Building Blocks	34
Lecture 8.....	39
Binary Numbers & Logic Operations	39
8.1 Why binary	42
8.2 Boolean Logic Operations.....	43
8.3 Truth Table for the XOR Operation	45
8.4 STRATEGY: Divide & Conquer.....	45
Lecture 9.....	47
HTML Lists & Tables (Web Development Lecture 3).....	47
9.1 Single Tags	47
9.2 Types of Lists	52
9.3 Ordered List Types.....	52
9.4 Useful URL	57
Lecture 10	59
Computer Software.....	59
10.1 Machine Language	59
10.2 Language Translators.....	59
10.3 Software Development.....	59
10.4 Major Types of SW	60
10.5 System SW are programs that	60
10.6 Operating System.....	60
10.7 Utilities:	61
10.8 Language Translators.....	61
10.9 Device Drivers.....	61
10.10 Application SW	61
10.11 Another way of classifying SW	62
10.12 Who Owns Software?	62
10.13 Main types of SW licensees	62
10.14 Proprietary SW License	62
10.15 Freeware SW License	63
10.16 Open-Source SW License	63
10.17 Shareware SW License	63
10.18 Trialware	63
Lecture 11.....	65
Operating Systems	65
11.1 Why Have OSes?	65
11.2 Core Tasks of an OS	65
11.3 OS Components	66

11.4 Kernel	67
11.5 Types of OS'es	67
11.6 Another Way of Classifying	67
11.7 How many different OS'es are there?	67
11.8 Comparing Popular OS'es	68
Lecture 12	69
Interactive Forms (Web Development Lecture 4)	69
12.1 Server-Side Scripts	71
12.2 Checkbox Input Element	76
12.3 Radio Button Input Element	77
12.4 Select from a (Drop Down) List	78
12.5 File Upload Input Element	79
Lecture 13	81
Application Software	81
13.1 Two Major Types of Software	81
13.2 Application Software	81
13.3 Classification According to the Mode	81
13.4 Classification According to Application Area	81
13.5 Scientific/Engineering/Graphics Apps	81
13.6 Scientific SW	82
13.7 Engineering SW	82
13.8 Graphics & Animation SW (1)	82
13.9 Business Applications	82
13.10 E-Commerce Software	82
13.11 ERP (Enterprise Resource Planning) SW	83
13.12 DSS (Decision Support Systems) SW	83
13.13 Productivity SW	83
13.14 Word Processors	83
13.15 Web Page Development SW	83
13.16 Spreadsheet SW (1)	83
13.17 Spreadsheet SW (2)	83
13.18 Presentation Development SW	83
13.19 Small-Scale Databases SW (1)	84
13.20 Small-Scale Databases SW (2)	84
13.21 Productivity SW Suites	84
13.22 Document-Centered Computing (DCC) - 1	84
13.23 Document-Centered Computing (DCC) - 2	84
13.24 Entertainment SW	84
13.25 Music & Video Players	84
13.26 Music Generation & Movie Editing SW	84
13.27 Games	84
13.28 Educational SW	85
13.29 Electronic Encyclopedias	85
13.30 On-Line Learning	85
13.31 Interactive CD's	85
13.32 Attributes of Good Application Software	85
13.33 Most Popular Application Software Categories	85
Lecture 14	87
Word Processing	87
14.1 Word Processor	87
14.2 Types: WYSIWYG-based & Markup-based	88
14.3 Desktop Publishing (DTP)	88

14.4 Word Processors for the Web	88
14.5 Let's try to use MS Word for creating a CV	90
Lecture 15	91
More on Interactive Forms (Web Development Lecture 5)	91
15.1 Single-Line Text Input Field	91
15.2 Password Input Field	91
15.3 Hidden Input	91
15.4 Checkbox Input Element	91
15.5 Radio Button Input Element	91
15.6 File Upload Input Element	91
15.7 Reset Button Input Element	92
15.8 Submit Button Input	92
15.9 Multi-Line Text Input Area	92
15.10 Select from a (Drop Down) List	92
15.11 Client-Side Scripting is a viable alternate	94
15.12 Server-Side Scripts: Review	94
15.13 Why JavaScript?	96
Lecture 16	99
Algorithms	99
16.1 Algorithm for Decimal-to-Binary Conversion	100
16.2 Algorithm (Better Definition)	100
16.3 Why Algorithms are Useful?	101
16.4 Analysis of Algorithms	101
16.5 Al-Khwarzmi	101
16.6 Greedy Algorithm	102
16.7 Deterministic Algorithm (1)	102
16.8 Randomized Algorithm (1)	102
16.9 Randomized Algorithm (2)	102
16.10 Deterministic Algorithm (2)	102
16.11 Heuristic	102
16.12 The Brute Force Strategy (1)	103
16.13 The Brute Force Strategy (2)	103
16.14 A Selection of Algorithmic Application Areas	103
16.15 Flowchart	104
Lecture 17	106
Algorithms II	106
17.1 Algorithm Building Blocks	106
17.2 Solution in Pseudo Code	108
17.3 Tips on Writing Good Pseudo Code	108
17.4 Pros and Cons of Flowcharts (1)	117
17.5 Pros and Cons of Flowcharts (2)	117
17.6 Pros and Cons of Pseudo Code (1)	117
17.7 Pros and Cons of Pseudo Code (2)	117
Lecture 18	118
Objects, Properties, Methods (Web Development Lecture 6)	118
18.1 New Concept: Client-Side Scripts	119
18.2 Advantages of Client-Side Scripting	119
18.3 Disadvantages	119
18.4 JavaScript	119
18.5 Client-Side JavaScript	120
18.6 Properties	121
18.7 Event Handlers	128

Lecture 19.....	129
Programming Languages.....	129
19.1 Batch Programs	129
19.2 Event-Driven Programs	129
19.3 Types of Prog. Languages.....	130
19.4 Programming SW Development	131
19.5 Object Oriented Design	131
19.6 Structured Design	131
19.7 Object-Oriented Languages	132
Lecture 20	133
SW Development Methodology	133
Lecture 21.....	142
Data Types & Operators (Web Development Lecture 7)	142
21.1 JavaScript Data Types.....	143
21.2 Declaring Variables.....	144
21.3 JavaScript Operators	148
21.4 Comparison Operators	148
21.5 Logical Operators	148
21.6 Elements of JavaScript Statements	149
Lecture 22	151
Spreadsheets.....	151
22.1 Business Plan for a New Software Development Company	151
22.2 The Structure of A Spreadsheet	152
22.3 Goal Seek	154
Lecture 23	158
Flow Control & Loops (Web Development Lecture 8).....	158
JavaScript Variables are Dynamically Typed	158
Lecture 24	166
Design Heuristics.....	166
24.1 Heuristic	166
24.2 System.....	166
24.3 System Architecture.....	166
24.4 Heuristics for system architecting.....	166
Lecture 25	170
Web Design for Usability	170
25.2 SPEED:	170
25.3 Elements of Website Design:	171
25.4 Website Navigation:	171
25.5 A Few Navigation Design Heuristics:	171
25.6 Navigation Design Heuristics (contd.):.....	171
25.7 Good designs assist the user in recovering from errors.....	173
25.8 Assisting the User Recover from Errors:.....	173
25.9 A few constructive recommendations	173
25.10 Making Display Elements Legible:.....	175
25.11 Ensuring Text is Readable:.....	175
25.12 Using Pictures & Illustrations:	176
25.13 Using Motion.....	176
Arrays (Web Development Lecture 9).....	177
26.1 Arrays in JavaScript.....	178
26.2 Array Identifiers.....	180
26.3 The 'length' Property of Arrays	181
26.4 Array Methods: sort() 26.5 Sorts the elements in alphabetical order .	181

26.6 Array Methods: reverse()	26.7 Reverses the order of the elements	181
26.7 Pseudo Code.....		182
Lecture 27		185
Computer Networks.....		185
27.1 Private Networks		186
27.2 Public Networks		186
27.3 VPN: Virtual Private Network (1)		187
27.4 Network Topologies		188
27.5 Networking Protocols		190
27.6 Types of Communication Channels		191
27.7 Network Security		191
Lecture 28		193
Introduction to the Internet		193
28.1 Internet: Network of Networks		196
28.2 Internet Networking Protocols		196
Lecture 29		199
Functions & Variable Scope (Web Development Lecture 10)		199
29.1 Function		199
29.2 Advantages of Functions		200
29.3 Function Identifiers		201
29.4 Arguments of a Function.....		201
29.5 Event Handlers.....		203
29.6 Scope of Variable		204
Lecture 30		209
Internet Services.....		209
30.1 Internet Addressing.....		210
30.2 DNS: Domain Name System.....		210
30.3 Internet Services		210
30.3 How does an eMail system work?.....		213
30.4 Using Instant Messaging		215
30.5 VoIP: Voice over IP.....		220
Lecture 31		221
Developing Presentations.....		221
31.1 Presentations:.....		221
31.2 The Structure of A Presentation:		224
31.3 Presentation Development SW:.....		224
Lecture 32		226
Event Handling (Web Development Lecture 11).....		226
32.1 What is Event Handling?		228
32.2 In-Line JavaScript Event Handling :		229
Lecture 33		234
Graphics & Animation		234
33.1 Computer Graphics:		235
33.2 Displaying Images:		235
33.3 Pixel Colors :.....		235
33.4 Color Mapping :.....		235
33.5 Dithering:		236
33.6 Aliasing:		236
33.7 Anti-Aliasing:.....		236
33.8 Graphics File Formats:.....		237
33.9 Vector or Object-Oriented Graphics:		237
33.10 Bit-Mapped or Raster Graphics:.....		237

33.11 File Formats Popular on the Web (1):.....	237
33.12 Image Processing:.....	237
33.13-D Graphics:	238
33.14 Animation:.....	238
Lecture 34	240
Intelligent Systems	240
34.1 (Artificial) Intelligent Systems:	241
34.2 Fuzzy Logic:	242
34.3 Robotics:.....	244
Lecture 35	245
Mathematical Methods (Web Development Lecture 12).....	245
35.1 Problems & Solutions:	246
35.2 Mathematical Functions in JavaScript:	248
Lecture 36	251
Data Management	251
36.1 BholiBooks.com :	252
36.2 Issues in Data Management:.....	252
36.3 DBMS :	253
36.4 OS Independence:	254
36.5 The Trouble with Flat-File Databases:	257
Lecture 37	259
Database Software	259
37.1 RDBMS.....	262
37.2 Some Terminology	263
Lecture 38	264
String Manipulations (Web Development Lecture 13)	264
38.1 String Manipulation in JavaScript.....	267
Lecture 39	274
Cyber Crime	274
39.1 07 February 2000	275
39.2 DoS Attack: A Cyber Crime.....	276
39.3 More cybercrimes	276
39.4 Viruses	277
39.5 Other Virus-Like Programs.....	278
Lecture 40	279
Social Implications of Computing	279
40.1 Introduction	280
40.2 Powerful Global Corporations	280
40.3 The Network Organization.....	280
40.4 Working from Home	281
40.5 From Mass- to Personalized-Marketing	281
40.6 The Political Process	282
Lecture 41.....	284
Images & Animation (Web Development Lecture 14)	284
41.1 Images in JavaScript	286
41.2 Flash Animation	293
Lecture 42	294
The Computing Profession	294
42.1 IT: Information Technology.....	295
42.2 Organization: A Collection of Teams	296
Lecture 43	302
The Future of Computing.....	302

Lecture 44	308
Programming Methodology (Web Development Lecture 15)	308
44.1 Design Guidelines	309
44.2 Coding Guidelines	309
44.3 Guidelines for Developing Short Programs	310
44.4 Design & Code Reviews	311
44.5 Testing & Debugging	312
44.6 Helpful Editors	314
Lecture 45	316
Review & Wrap-Up	316
Course Objectives	323

Lecture 1

Introduction

1.1. Charles Babbage (1791-1871)

Creator of the Analytical Engine - the first general-purpose digital computer (1833)
The Analytical Engine was not built until 1943 (in the form of the Harvard Mark I)

1.2. The Analytical Engine

A programmable, mechanical, digital machine
Could carryout any calculation
Could make decisions based upon the results of the previous calculation
Components: input; memory; processor; output

1.3. Ada, Countess of Lovelace(1815-52)

Babbage: the father of computing

Ada: the mother?

Wrote a program for computing the Bernoulli's sequence on the Analytical Engine - world's 1st computer program

Ada: A programming language specifically designed by the US Dept of Defense for developing military applications was named Ada to honor her contributions towards computing

A lesson that we all can learn from Babbage's Life

Charles Babbage had huge difficulties raising money to fund his research

As a last resort, he designed a clever mathematical scheme along with Ada, the Countess of Lovelace

It was designed to increase their odds while gambling. They bet money on horse races to raise enough money to support their research experiments

Guess what happened at the end? They lost every penny that they had.

Fast

Bored

Storage

Here is a fact:



It could analyze up to 300 billion chess moves in three minutes

In 1997 Deep Blue, a supercomputer designed by IBM, beat Gary Kasparov, the World Chess Champion

That computer was exceptionally fast, did not get tired or bored. It just kept on **analyzing** the situation and kept on **searching** until it found the perfect move from its list of possible moves ...

Goals for Today:

To develop an appreciation about the capabilities of computing

To find about the structure & policies of this course

CS101 Introduction to Computing

1.4 Course Contents & Structure

Course Objectives

To build an appreciation for the fundamental concepts in computing

To achieve a beginners proficiency in Web page development

To become familiar with popular PC productivity software

Week	Lecture 1	Lecture 2	Lecture 3 Web Dev	Readings		Assignment
				UC	JS	
1						
2						
3						
4						
5						
6						
7						
8	Midterm Exam					
9						
10						
11						
12						
13						
14						
15						
	Finals Week					

Fundamental concepts

Intro to computing

Evolution of computing

Computer organization

Building a PC

Microprocessors

Binary numbers & logic

Computer software

Operating systems

Application software

Algorithms

Flowcharts

Programming languages

Development methodology

Design heuristics

Web design for usability

Computer networks

Intro to the Internet

Internet services

Graphics & animation

Intelligent systems

Data management

Cyber crime

Social implications

The computing profession

The future of computing

Web page development**Web Development**

The World Wide Web
Making a Web page
Lists & tables
Interactive forms
Objective & methods
Data types & operators

Flow control & loops Arrays
Built-in functions
User-defined functions
Events handling
String manipulation
Images & graphics
Programming methodology

Productivity Applications

Word processor
Spreadsheet
Presentation software
Database

Instructor:

Altaf Khan
cs101@vu.edu.pk

Course Web Page:

<http://vulms.vu.edu.pk/>

Textbooks:

UC - Understanding Computers (2000 ed.)
JS - Learn JavaScript in a Weekend

Reading Assignments

Please make sure to read the assigned material for each week before the commencement of the corresponding week

Reading that material beforehand will help you greatly in absorbing with ease the matter discussed during the lecture

Check your e-mail often for announcements related to this and other VU courses

Marks

distribution ...

Assignments (15%)

Almost one every week, 13 in all
No credit for late submissions
The lowest 2 assignment grades will be dropped

Midterm Exam (35%)

During the 8th week
Duration: One hour
Will cover all material covered during the first seven weeks

Final Exam (50%)

During the 16th week
Will cover the whole of the course with a slight emphasis on the material covered after the midterm exam

Duration: 2 hours

First Assignment

Send an email message to me at altaf@vu.edu.pk with the subject “Assignment 1” giving me some information (in around 50 words) about what you see yourself doing ten years from now

Go to the CS101 message board and post a message (consisting of approx. 50 words) about how we could make the contents of this course more suitable for your individual needs. The subject for this message should be “Assignment 1”

Consult the CS101 syllabus for the submission deadline

A suggestion about unfamiliar terms

We try not to use any new terms without explaining them first

However, it is not possible to do that all the time

If you encounter any unfamiliar terms during the lectures, please note them down and consult the GLOSSARY provided at the end of the “Understanding Computers” text book for their meaning

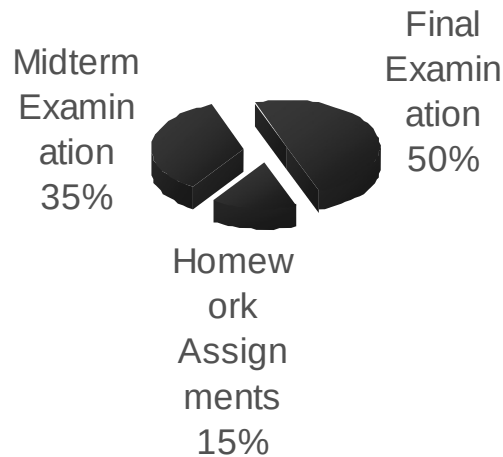
Let’s summarize the things that we have covered today?

A few things about:

the very first digital computer & its inventor

the capability of modern computers

the structure and contents of CS101



In the Next Lecture ...

We’ll continue the story of the evolution of digital computers from the Analytical Engine onwards.

We’ll discuss many of the key inventions and developments that have led to the shape of the current field of computing.

Lecture 2

Evolution of Computing

Today's Goal

To learn about the evolution of computing

To recount the important and key events

To identify some of the milestones in computer development

Babbage's Analytical Engine - 1833

Mechanical, digital, general-purpose

Was crank-driven

Could store instructions

Could perform mathematical calculations

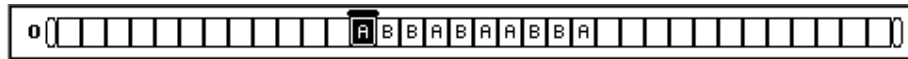
Had the ability to print

Could punched cards as permanent memory

Invented by Joseph-Marie Jacquard

2.1 Turing Machine – 1936

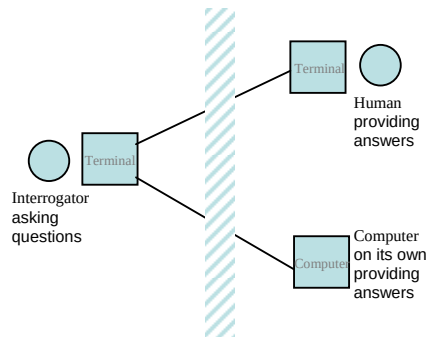
Introduced by Alan Turing in 1936, Turing machines are one of the key abstractions used in modern computability theory, the study of what computers can and cannot do. A Turing machine is a particularly simple kind of computer, one whose operations are limited to reading and writing symbols on a tape, or moving along the tape to the left or right. The tape is marked off into squares, each of which can be filled with at most one symbol. At any given point in its operation, the Turing machine can only read or write on one of these squares, the square located directly below its "read/write" head.



2.2 The “Turing test”

A test proposed to determine if a computer has the ability to think. In 1950, Alan Turing (Turing, 1950) proposed a method for determining if machines can think. This method is known as The Turing Test.

The test is conducted with two people and a machine. One person plays the role of an interrogator and is in a separate room from the machine and the other person. The interrogator only knows the person and machine as A and B. The interrogator does not know which the person is and which the machine is. Using a teletype, the interrogator, can ask A and B any question he/she wishes. The aim of the interrogator is to determine which the person is and which the machine is. The aim of the machine is to fool the interrogator into thinking that it is a person. If the machine succeeds then we can conclude that machines can think.



2.3 Vacuum Tube – 1904:

A vacuum tube is just that: a glass tube surrounding a vacuum (an area from which all gases has been removed). What makes it interesting is that when electrical contacts are put on the ends, you can get a current to flow though that vacuum.

A British scientist named John A. Fleming made a vacuum tube known today as a diode. Then the diode was known as a "valve,"

2.4 ABC – 1939

The Atanasoff-Berry Computer was the world's first electronic digital computer. It was built by John Vincent Atanasoff and Clifford Berry at Iowa State University during 1937-42. It incorporated several major innovations in computing including the use of binary arithmetic, regenerative memory, parallel processing, and separation of memory and computing functions.

2.5 Harvard Mark 1 – 1943:

Howard Aiken and Grace Hopper designed the MARK series of computers at Harvard University. The MARK series of computers began with the Mark I in 1944. Imagine a giant roomful of noisy, clicking metal parts, 55 feet long and 8 feet high. The 5-ton device contained almost 760,000 separate pieces. Used by the US Navy for gunnery and ballistic calculations, the Mark I was in operation until 1959.

The computer, controlled by pre-punched paper tape, could carry out addition, subtraction, multiplication, division and reference to previous results. It had special subroutines for logarithms and trigonometric functions and used 23 decimal place numbers. Data was stored and counted mechanically using 3000 decimal storage wheels, 1400 rotary dial switches, and 500 miles of wire. Its electromagnetic relays classified the machine as a relay computer. All output was displayed on an electric typewriter. By today's standards, the Mark I was slow, requiring 3-5 seconds for a multiplication operation

2.6 ENIAC – 1946:

ENIAC I (**E**lectrical **N**umerical **I**ntegrator **A**nd **C**alculator). The U.S. military sponsored their research; they needed a calculating device for writing artillery-firing tables (the settings used for different weapons under varied conditions for target accuracy).

John Mauchly was the chief consultant and J Presper Eckert was the chief engineer. Eckert was a graduate student studying at the Moore School when he met John Mauchly in 1943. It took the team about one year to design the ENIAC and 18 months and 500,000 tax dollars to build it.

The ENIAC contained 17,468 vacuum tubes, along with 70,000 resistors and 10,000 capacitors.

2.7 Transistor – 1947

The first transistor was invented at Bell Laboratories on December 16, 1947 by William Shockley. This was perhaps the most important electronics event of the 20th century, as it later made possible the integrated circuit and microprocessor that are the basis of modern electronics. Prior to the transistor the only alternative to its current regulation and switching functions (TRANSfer resISTOR) was the vacuum tubes, which could only be miniaturized to a certain extent, and wasted a lot of energy in the form of heat.

Compared to vacuum tubes, it offered:

smaller size

better reliability

lower power consumption

lower cost

2.8 Floppy Disk – 1950

Invented at the Imperial University in Tokyo by Yoshiro Nakamats

2.9 UNIVAC 1 – 1951

UNIVAC-1. The first commercially successful electronic computer, UNIVAC I, was also the first general purpose computer - designed to handle both numeric and textual information. It was designed by J. Presper Eckert and John Mauchly. The implementation of this machine marked the real beginning of the computer era.

Remington Rand delivered the first UNIVAC machine to the U.S. Bureau of Census in 1951. This machine used magnetic tape for input.

first successful commercial computer

design was derived from the ENIAC (same developers)

first client = U.S. Bureau of the Census

\$1 million

48 systems built

2.10 Compiler - 1952

Grace Murray Hopper an employee of Remington-Rand worked on the UNIVAC. She took up the concept of reusable software in her 1952 paper entitled "The Education of a Computer" and developed the first software that could translate symbols of higher computer languages into machine language. (Compiler)

2.11 ARPANET – 1969

The Advanced Research Projects Agency was formed with an emphasis towards research, and thus was not oriented only to a military product. The formation of this agency was part of the U.S. reaction to the then Soviet Union's launch of Sputnik in 1957. (ARPA draft, III-6). ARPA was assigned to research how to utilize their investment in computers via Command and Control Research (CCR). Dr. J.C.R. Licklider was chosen to head this effort.

Developed for the US DoD Advanced Research Projects Agency

60,000 computers connected for communication among research organizations and universities

2.12 Intel 4004 – 1971

The 4004 was the world's first universal microprocessor. In the late 1960s, many scientists had discussed the possibility of a computer on a chip, but nearly everyone felt that integrated circuit technology was not yet ready to support such a chip. Intel's Ted Hoff felt differently; he was the first person to recognize that the new silicon-gated MOS technology might make a single-chip CPU (central processing unit) possible.

Hoff and the Intel team developed such architecture with just over 2,300 transistors in an area of only 3 by 4 millimeters. With its 4-bit CPU, command register, decoder, decoding control, control monitoring of machine commands and interim register, the 4004 was one heck of a little invention. Today's 64-bit microprocessors are still based on similar designs, and the microprocessor is still the most complex mass-produced product ever with more than 5.5 million transistors performing hundreds of millions of calculations each second - numbers that are sure to be outdated fast.

2.13 Altair 8800 – 1975

By 1975 the market for the personal computer was demanding a product that did not require an electrical engineering background and thus the first mass produced and marketed personal computer (available both as a kit or assembled) was welcomed with open arms. Developers Edward Roberts, William Yates and Jim Bybee spent 1973-1974 to develop the MITS (Micro Instruments Telemetry Systems) Altair 8800. The price was \$375, contained 256 bytes of memory (not 256k), but had no keyboard, no display, and no auxiliary storage device. Later, Bill Gates and Paul Allen wrote their first product for the Altair -- a BASIC compiler (named after a planet on a *Star Trek* episode).

